

# SYNTH-OEM-A diagnostic specification

OEM OEM-A | version 1.0

| Data identifiers |       |       |               |                |
|------------------|-------|-------|---------------|----------------|
| DID              | Name  | Bytes | Read sessions | Write sessions |
| 0x1000           | DID-0 | 4     | 1, 3          | 3              |
| 0x1001           | DID-1 | 4     | 1, 3          | 3              |
| 0x1002           | DID-2 | 4     | 1, 3          | 3              |
| 0x1003           | DID-3 | 4     | 1, 3          | 3              |
| 0x1004           | DID-4 | 4     | 1, 3          | 3              |

| Data identifiers |       |       |               |                |
|------------------|-------|-------|---------------|----------------|
| DID              | Name  | Bytes | Read sessions | Write sessions |
| 0x1005           | DID-5 | 4     | 1, 3          | 3              |
| 0x1006           | DID-6 | 4     | 1, 3          | 3              |
| 0x1007           | DID-7 | 4     | 1, 3          | 3              |
| 0x1008           | DID-8 | 4     | 1, 3          | 3              |
| 0x1009           | DID-9 | 4     | 1, 3          | 3              |

## Prose appendix: operating conditions

DID 0x1000: signal\_0 is true.

DID 0x1001: signal\_1 is true.

DID 0x1002: signal\_2 is true.

DID 0x1003: signal\_3 is true.

DID 0x1004: signal\_4 is true.

DID 0x1005: signal\_5 is true.

The six documented operating conditions are signal\_0 through signal\_5; each must be true before its corresponding write is accepted.

sessions: [{ 'id': 1, 'name': 'default'}, { 'id': 3, 'name': 'extended'}]

security\_levels: [{ 'level': 0, 'name': 'locked'}, { 'level': 1, 'name': 'level1'}]

services: [{ 'sid': 16, 'name': 'SessionControl'}, { 'sid': 17, 'name': 'Reset', 'subfunctions': [{ 'value': 1, 'name': 'hard'}]}, { 'sid': 34, 'name': 'ReadDataByIdentifier'}, { 'sid': 39, 'name': 'SecurityAccess'}, { 'sid': 46, 'name': 'WriteDataByIdentifier'}, { 'sid': 49, 'name': 'RoutineControl', 'allowed\_sessions': [3], 'required\_security\_level': 1}, { 'sid': 62, 'name': 'TesterPresent', 'suppress\_positive\_response\_supported': True}]

routines: [{ 'rid': 8192, 'name': 'Routine-0', 'control\_types': [1, 2, 3], 'sessions': [3], 'security': 1, 'parameter\_lengths': { '1': 1, '2': 0, '3': 2}}, { 'rid': 8193, 'name': 'Routine-1', 'control\_types': [1, 2, 3], 'sessions': [3], 'security': 1, 'parameter\_lengths': { '1': 1, '2': 0, '3': 2}}, { 'rid': 8194, 'name': 'Routine-2', 'control\_types': [1, 2, 3], 'sessions': [3], 'security': 1, 'parameter\_lengths': { '1': 1, '2': 0, '3': 2}}]

timing: { 'p2\_ms': 50, 'p2\_star\_ms': 5000, 's3\_ms': 5000}

nrc\_priority: [19, 18, 17, 127, 126, 51, 53, 36, 49, 34]